

Chapter 1

Basic Definitions

Definition 1.1 (Connectedness). A metric space X is connected if and only if the only subsets of X which are open and closed are X and $\emptyset$.

Definition 1.2 (Disconnection). A disconnection of a metric space is a pair of open sets $U, V \in \mathcal{T}/\{\emptyset\}$ where $U \cap V = \emptyset$ and $X = U \cup V$

Thus X is connected $\Leftrightarrow X$ has no disconnection.

Proposition 1.3. Let X be a metric space and let $\{A_j\}_{j \in J}$ be a family of connected subsets of X , such that for $(j', j'') \in J \times J$ there exists $\{j_k\}_{k=1}^N \subset J$ where $j_1 = j'$ and $j_N = j''$ where $A_{j_k} \cap A_{j_{k+1}} \neq \emptyset$ then

$$A = \bigcup_{k=1}^N A_{j_k}$$

is connected.

Chapter 2

Main Theorem

Theorem 2.1. A subset of $\mathbb{R}$ is connected if and only if its an interval

Proof: $\Rightarrow$: Let $C \subseteq \mathbb{R}$ with at least 2 distinct elements $a < b$. If $x \in (a, b)$ such that $x \notin C$. Then $(-\infty, x) \cap C$, $C \cap (x, \infty)$ is a disconnection. Thus C has the property that if $a, b \in C$ where $a < b$ then $(a, b) \subseteq C$. This means that C is a interval.

$\Leftarrow$: Let $I \subset \mathbb{R}$ be a nonempty interval. Assume there is a disconnection $U_1, U_2 \in \mathbb{R}$ such that $U_j \cap I \neq \emptyset$ and $I \subseteq U_1 \cup U_2$. Let $a_j \in I \cap U_j$ we may assume $a_1 < a_2$. Let

$$b = \sup([a_1, a_2] \cap U_1) \Rightarrow a_1 \leq b \leq a_2 \Rightarrow b \in [a_1, a_2] \subseteq I.$$

If $b \in U_1$, then $a_1 \leq b < a_2$. Take $\exists \varepsilon_1 > 0$ such that $b + \varepsilon_1 < a_2$ and $b + \varepsilon_1 \in U_1$ this contradicts the supremum nature of b . Thus $b \in U_2$, then $a_1 < b \leq a_2$. There exists a $\varepsilon_2 > 0$ such that $[b_1 - \varepsilon, b_1 + \varepsilon] \subseteq U_2 \cap (a_1, \infty)$. By the definition of least upper bound, $b - \varepsilon_2$ is not an

upper bound for $[a_1, a_2] \cap U_1$ so there exists $c \in [a_1, a_2] \cap U_1$ where $a_1 < b - \varepsilon_2 < c \leq b \leq a_2$. Thus $c \in U_1 \cap U_2$ but impossible because $U_1 \cap U_2 = \emptyset$. $\square$

Chapter 3

Continuity Method:

Let $I \subset \mathbb{R}$ be an interval and $0 \leq y \in C(I)$ be such that $y(t) \leq A + \varepsilon F(y(t))$ for $t \in I$ and $A > 0, \varepsilon > 0, F : [0, \infty) \rightarrow [0, \infty)$ is locally bounded:

$$\sup_{\xi \in [0, R]} F(\xi) < \infty, \forall R > 0$$

Assume also that $y(t_0) \leq 2A$ for some $t_0 \in I$.

Proposition 3.1. For $\varepsilon > 0$ small enough, $y(t) \leq 2A$ for all $t \in I$.

Proof: Let $E = \{t \in I : y(t) \leq 2A\} \neq \emptyset$ because $t_0 \in E$. E is closed (y is continuous). We will show for $\varepsilon > 0$ small enough E is open and hence a interval. Let $t_1 \in E$, y continuous thus $\exists \delta > 0$ so that $t \in (t_1 - \delta, t_1 + \delta)$ so $|y(t) - y(t_1)| < 1$. We get for $t \in (t_1 - \delta, t_1 + \delta)$. $y(t) \leq A + \varepsilon F(y(t))$. We know $0 \leq y(t) \leq y(t_1) + 1 \leq 2A + 1$.

Thus

$$y(t) \leq A + \varepsilon \sup_{\xi \in [0, 2A+1]} F(\xi).$$

Thus for $\varepsilon > 0$ small enough $y(t) \leq 2A$. Thus E is open. Thus $E = I$. $\square$

Chapter 3

References